

Bounded and Auditable Progression Synthesis for Brain Metastases: A Lightweight Two-Stage Latent Diffusion Framework with a Generation-Isolation Audit

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“What might this lesion look like next?” is treated as a bounded, provable and auditable synthesis problem — not as a claim of clinical forecasting.

1 Motivation

Brain metastases are the most common intracranial secondary tumours, and therapy decisions hinge on how lesions change over follow-up. Routine reading answers “what is there now”, not “what comes next”, while multi-timepoint scans of the same patient are scarce — so “future morphology” has lacked a testable, quantitative handle. Existing 3D brain-tumour synthesis works synthesize inside a segmentation mask with static conditioning, and assume a pretrained backbone and large-scale compute — hard to reproduce on one machine.

2 Method (Fig. 1)

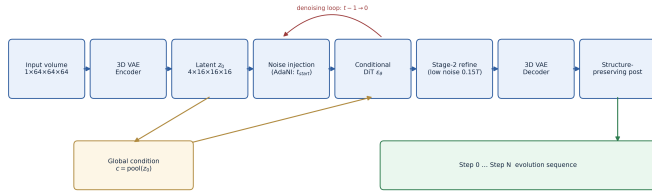


Fig. 1 Self-contained two-stage latent-diffusion pipeline (schematic).

All components are custom-trained; no external pretrained backbone is used. A 3D convolutional autoencoder maps a $64 \times 64 \times 64$ T1-Gd volume to a $4 \times 16 \times 16 \times 16$ latent. A Diffusion Transformer (dim 64, 2 blocks, 4 heads, patch $2 \times 2 \times 2$, 512 latent tokens; $T=200$ linear β from 1×10^{-4} to 0.02) predicts latent noise conditioned on a globally pooled latent. Stage 1 selects the starting noise level from the prescribed step span (AdaNI); stage 2 re-noises to $0.15T$ and refines texture. A structure-preserving post-process locks the brain contour and confines the residual to a lesion-candidate mask, with symmetric clipping at $c=0.08$ and strength $s=0.35$.

3 Growth-law time weighting replaces uniform interpolation

Weights between the two endpoints follow the normalised volume progress of an explicit growth law (exponential / Gompertz / logistic-Fisher-KPP), parameterised by an equivalent volume-doubling time T_d . Five propositions are stated and machine-checked on all 51 test cases: load-weight affinity; the convex-combination identity; the exponential adjacent-increment bound L ; the closed-form maximiser t^* of the deviation from linear weights; and trajectory boundedness, $|x(t) - x_0| = w(t) \cdot |x_N - x_0|$, with $|A_{\text{gen}}| \leq s \cdot c = 0.028$. Measured: $|x_N - x_0|_\infty = 0.1872 \pm 0.0195$; deviation from linear weights 0.186505824 vs the closed form 0.186505833 (rel. 4.4×10^{-8}).

4 Results: structure preservation (Fig. 2)

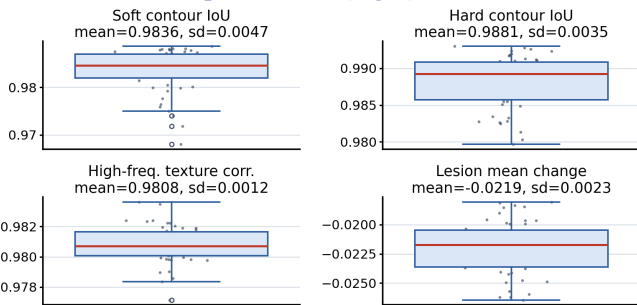


Fig. 2 Progression-simulation metrics over the 51-case test set — BrainMetShare, 156 studies, 105 labelled training / 51 test cases (the publisher’s own split); $N = 100$ steps, box = IQR, dots = per-case.

Soft contour IoU 0.9836 ± 0.0047 , hard IoU 0.9881 ± 0.0035 , high-frequency texture correlation 0.9808 ± 0.0012 , lesion-region mean intensity change -0.0219 ± 0.0023 . Change stays inside the candidate neighbourhood (outside/inside absolute-change ratio 0.633, range 0.416–0.863) but is *not* concentrated inside the lesion.

5 Qualitative progression (Fig. 3)

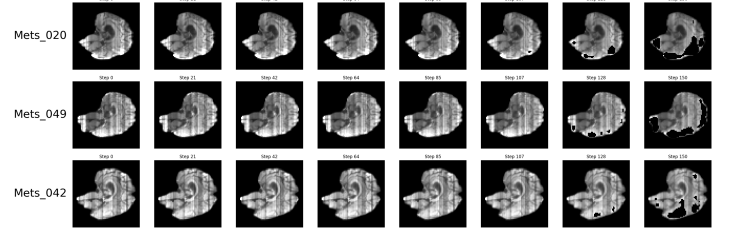


Fig. 3 Test cases Mets_020 / 049 / 042, Step 0→150 (axial; a denser grid than the statistics above, for display only).

6 Generation-isolation audit (Fig. 4) — the decisive check

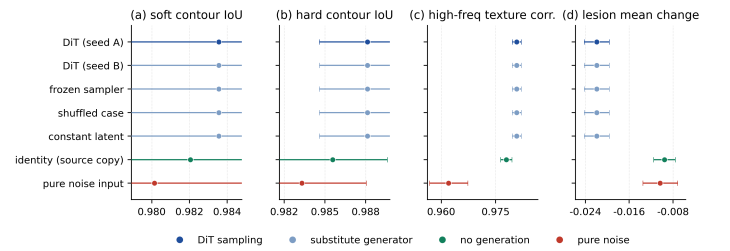


Fig. 4 Replacing the sampler with surrogates leaves the reported metrics essentially unchanged.

Replacing the entire sampler by frozen sampling, a constant latent, or another patient’s result leaves all four metrics bit-identical; a source copy (“no generation”) and pure noise land in the same band (per-case ranges $0.0034 / 0.0048 / 0.0188 / 0.0129$). Root cause: the decoder’s final Sigmoid saturates — mean latent magnitude 11.43, and 99.97% of candidate-core voxels exceed the 0.08 clip — so the reported numbers characterise the deterministic post-processing chain, not the sampler.

7 The root cause is repairable — but not sufficient

Two changes only (EMA latent standardisation; Hardtanh(0,1) decoder output) under an otherwise identical recipe (3 VAE + 5 DiT epochs, same loss, optimiser and seed; 4.5 min on one machine) give: reconstruction PSNR $15.993 \pm 1.497 \rightarrow 29.299 \pm 1.318$ dB; decoder output std $0.0010 \rightarrow 0.1447$; candidate-core clipping saturation (frozen sampling) $99.97\% \rightarrow 34.98\%$; post-processed core surrogate difference $0.000 \rightarrow 0.0433$. The sampler’s change, however, remains non-localised (core/ring residual contrast 0.990, versus 1.412 for a plain autoencoder round-trip and 1.706 for a random latent), so the repair restores a non-degenerate decoder without establishing lesion-specific generation.

8 Take-home and limits

We contribute a reproducible, self-contained LDM together with a reusable isolation-audit protocol and five machine-checkable propositions — i.e. change-bounded, structure-verifiable and bit-reproducible controllable synthesis. Limits: no longitudinal ground truth exists here (the step index is a generation condition, not clinical time); T_d is not calibrated on any cohort; and the heuristic candidate mask is far larger than expert lesions (24.39% vs 0.20% of brain volume, Dice 0.0071), so mask-defined numbers must not be read as lesion-internal quantification. Every figure and number above is regenerated by the released scripts.

Reproducibility

Code, checkpoints and every script that produced the numbers above are released. A paper-to-code consistency checker re-derives all 220 quoted quantitative claims from the raw outputs (220 pass / 0 fail); a full 51-case recompute takes ≈ 31 s on CPU (4.7 s reconstruction + 26.1 s simulation) and is bit-reproducible for a fixed weight set and seed, while CPU and MPS agree to four significant digits.